

COMPLETE GUIDE

CRM Project — Full Architecture & Module Guide

A comprehensive reference covering system flow, module design, data models, implementation phases, and best practices for building a production-grade Customer Relationship Management platform.

11
Core Modules

5
Build Phases

360°
Customer View

100%
Documented

01 What is a CRM System?

Foundations, purpose, and why it matters

A Customer Relationship Management (CRM) system is a centralized platform that manages all interactions between a business and its customers — from the first marketing touchpoint to post-sale support. It replaces disconnected spreadsheets, email threads, and siloed tools with a single source of truth.

Core Goals

- Centralize all customer and prospect data
- Track the entire sales pipeline end-to-end
- Automate repetitive sales and marketing tasks
- Give teams visibility into customer history
- Generate reports that drive informed decisions
- Improve customer retention and satisfaction

Who Uses It?

Role	Primary Use
Sales Reps	Manage leads & deals
Managers	Pipeline oversight & forecasting
Marketing	Campaign tracking & lead generation
Support	Ticket mgmt & history
Executives	Dashboards & KPI reports

02 System Flow — How It Works

End-to-end journey from lead capture to closed deal

1	Lead Capture & Entry	Leads enter the CRM via direct input or third-party integrations.
2	Lead Qualification	Raw leads are evaluated against criteria like firmographics, behavior, and intent. This stage often involves "Marketing Qualified Lead" (MQL) designation.
3	Lead Assignment & Ownership	Qualified leads are assigned to a sales rep or team. This assignment triggers a notification to the responsible party.
4	Lead → Opportunity Conversion	When a rep confirms a lead is ready for sales, it is converted into an Opportunity. All previous lead history is preserved.
5	Sales Pipeline Progression	The Opportunity moves through the sales pipeline stages (e.g., Discovery, Proposal, Negotiation). The rep updates the status. The Opportunity is marked as Won/Lost). Each stage has associated tasks and deadlines.
6	Close & Post-Sale Handoff	On "Closed Won," the deal is finalized. The account is handed off to the support or customer success team for onboarding, training, and initial support.
7	Ongoing Customer Relationship	Post-close, the Account Manager (AM) or Customer Success Manager (CSM) maintains the relationship health, identifies upsell/cross-sell opportunities, and ensures customer satisfaction.

03 Lead Lifecycle Statuses

All possible states a record can be in

Status	Meaning	Next Action	Owner
New	Just entered, not yet reviewed	Assign to rep or SDR	System/Admin
Attempted	Rep tried to contact, no response	Follow-up in 2 days	Sales Rep
Connected	Initial conversation happened	Schedule discovery call	Sales Rep
MQL	Meets scoring threshold	Pass to sales	Marketing
SQL	Rep confirmed fit	Convert to Opportunity	Sales Rep
Nurturing	Not ready — in drip sequence	Re-engage on trigger	Automation
Disqualified	Not a fit	Archive + tag reason	Sales Rep
Converted	Became Contact + Account + Opp	Begin sales process	Sales Rep

04 Contacts & Accounts Module

The foundational data objects of any CRM

Contacts Module

A Contact represents an individual person — a customer, prospect, or stakeholder. Every interaction is logged against their record, giving a complete 360° history.

- Full name, title, email, phone (multiple), social profiles
- Linked to parent Account (company)
- Activity timeline: calls, emails, meetings, notes
- Custom fields for industry-specific data
- Contact stage / relationship status
- Tags and segments for bulk actions
- Email opt-in / GDPR consent tracking

Accounts Module

An Account is a company or organization. It can have many Contacts, multiple Opportunities, and a full billing and support history.

- Company name, website, industry, size, revenue
- Parent-child account hierarchy (subsidiaries)
- All associated Contacts listed with roles
- All Opportunities past and present
- Support tickets and SLA status
- Account health score (composite metric)
- Assigned Account Executive / CSM
- Renewal dates and contract details

05 Leads & Opportunities

The core sales objects driving pipeline

Leads Module

Unqualified prospects kept separate from Contacts to avoid polluting your clean customer database.

- Lead source tracking (web, referral, event, ad)
- Lead score (auto-calculated by rules engine)
- Assigned rep + reassignment history
- Status workflow: New → Contacted → Qualified → Converted
- Web tracking: page visits, email opens
- Bulk assignment and status update

Opportunities Module

An Opportunity (or Deal) represents a real sales engagement with a defined value and close date.

- Deal amount, expected close date, probability %
- Pipeline stage with visual Kanban board
- Products / line items linked to the deal
- Competitors tracked on the deal
- Win/loss reason mandatory on close
- Forecast category (Commit / Best Case / Pipeline)
- Linked quotes and proposals

06 Sales Pipeline Module

Visual pipeline management and forecasting

Stage	Definition	Entry Criteria	Probability
Prospecting	Initial outreach made	Lead converted	10%
Discovery	Needs assessment call completed	Call logged with notes	20%
Proposal	Formal proposal delivered	Proposal document attached	40%
Negotiation	Pricing/terms being discussed	Proposal viewed by prospect	60%
Contract Sent	Agreement sent for signature	Contract uploaded	80%
Closed Won	Deal signed & confirmed	Signed contract + PO	100%
Closed Lost	Lost or no decision	Loss reason selected	0%

Pipeline Features: Kanban view (drag deals between stages), List view, Forecast view. Managers see team-level pipeline; reps see their own. Filters by date range, rep, region, product, or deal size.

07 Activities & Tasks Module

Track every touchpoint and keep reps accountable

Calls

Log inbound/outbound calls with duration, outcome, and notes. Integrates with VoIP systems.

- Auto-log from integrated dialer
- Call disposition (Connected, No Answer, Voicemail)
- Follow-up task auto-created

Meetings

Schedule and track meetings synced with Google Calendar / Outlook.

- Two-way calendar sync
- Post-meeting notes and action items
- Video conference link auto-generated

Tasks

Reps manage daily follow-ups, reminders, and assigned action items.

- Due date, priority, and assigned user
- Linked to Contact / Account / Deal
- Overdue task alerts

Notes

Rich-text notes visible to all team members with access to the record.

- Rich text with @mentions
- Pinned notes for key information
- Searchable across all records

08 Communications Module

Unified inbox and email tracking

Email Integration

- Two-way Gmail / Outlook sync
- Email open & link-click tracking
- Email templates with merge fields
- Sequences (automated multi-step outreach)
- BCC-to-CRM email logging
- Shared team inbox for inbound

Other Channels

- SMS / WhatsApp via Twilio
- Live chat widget logged to CRM record
- Social media mentions tracked
- VoIP calls auto-logged
- Video meeting recordings linked

09 Reports & Analytics Module

Dashboards and insights that drive decisions

Sales Dashboards

- Total pipeline value by rep / team / region
- Monthly & quarterly revenue vs. target
- Win rate % (overall and by stage)
- Average deal size and sales cycle length

Lead Analytics

- Lead volume by source (web, referral, ads)
- Lead-to-MQL and MQL-to-SQL conversion rates

- Time to first contact (response time SLA)
- Funnel drop-off analysis by stage

Forecast Reports

- Weighted pipeline forecast (amount × probability)
- Commit vs. Best Case vs. Pipeline categories
- Quarter close projection with trend lines
- Rep-level forecast vs. quota

Activity Reports

- Calls made / emails sent per rep per day
- Tasks completed vs. overdue
- Meeting-to-proposal conversion rate
- Leaderboard view for team motivation

Custom Report Builder: Drag-and-drop editor — choose object, select fields, apply filters, group by dimension, choose visualization (bar, line, pie, table). Reports can be scheduled to email as PDF/CSV.

10 Automation & Workflows Module

Eliminate manual work and enforce process consistency

Workflows are trigger → condition → action chains that run automatically. They enforce your process, reduce manual work, and ensure nothing falls through the cracks.

Trigger	Condition	Action	Example Use
Lead created	Source = Website	Assign to SDR team	Inbound routing
Lead score ≥ 80	Always	Create task for rep	Hot lead alert
Deal stage changed	Stage = Proposal	Email proposal template	Auto follow-up
Task overdue 24h	Always	Notify manager	Accountability
Deal closed won	Always	Create onboarding tasks	Handoff workflow
Contact inactive 30d	Stage = Customer	Trigger re-engagement email	Retention
Renewal date -90d	Type = Enterprise	Create renewal opportunity	Renewal automation

11 Marketing Module

Campaigns, email sequences, and attribution

Email Campaigns

- Drag-and-drop email builder
- Segment-based sending
- A/B test subject lines
- Open, click, bounce, unsubscribe tracking
- Campaign-to-revenue attribution
- GDPR / CAN-SPAM compliance tools

Lead Nurturing

- Multi-step drip email sequences
- Behavior-triggered branching logic
- Content download tracking as lead events
- Webinar / event registration integration
- Landing page form connector
- UTM parameter auto-capture

12 Customer Support Module

Ticketing, SLAs, and satisfaction tracking

Ticketing

Support tickets linked to Contact and Account so any rep sees open issues before a call.

- Ticket creation via email, form, phone
- Priority tiers (Low / Medium / High / Critical)
- SLA clock per priority tier
- Escalation rules if SLA breached
- Internal notes vs. customer-visible replies

CSAT / NPS

Auto-send satisfaction surveys after ticket resolution or at regular intervals.

- CSAT survey sent on ticket close
- Quarterly NPS survey automation
- Responses logged to Account record
- Detractor alert triggers immediate follow-up task

13 User & Role Management

Access control and team structure

Role	Access Level	Can See	Can Edit
Admin	Full	All records, settings, billing	Everything incl. system config
Sales Manager	Team	Own team's all records	Team deals, assign leads, reports
Sales Rep	Own+TeamView	Own leads/deals + read team	Own assigned records
Marketing	Functional	Leads, campaigns, contacts	Marketing fields, campaigns
Support Agent	Functional	Tickets, contacts, accounts	Support tickets
Read Only	Limited	Reports and dashboards only	Nothing

Field-Level Security: Beyond role-level access, individual fields (e.g. contract value, personal data) can be hidden from specific roles. Critical for GDPR compliance.

14 Integrations

Connecting the CRM to your full tech stack

Category	Tools	Integration Type	Data Flowing
Email	Gmail, Outlook, Exchange	OAuth / API	Emails, contacts, calendar
VoIP	Twilio, RingCentral, Aircall	Native / Webhook	Call logs, recordings, SMS
Marketing	Mailchimp, ActiveCampaign	API Sync	Contacts, campaigns, opt-outs
ERP/Billing	QuickBooks, Stripe	Webhook + API	Invoices, payments, subs
Support	Intercom, Zendesk, Freshdesk	API	Tickets, chat logs, CSAT
Meetings	Zoom, Calendly, Google Meet	OAuth	Meeting links, recordings
Enrichment	Clearbit, ZoomInfo, Apollo	API / Webhook	Company data, contact info
Automation	Zapier, Make (Integromat)	Webhook / REST	Bidirectional any-field sync
Analytics	Google Analytics, Mixpanel	JS snippet + API	Web events, funnel data

15 Recommended Tech Stack

Building a CRM from scratch — sensible defaults

React / Next.js Frontend UI	Node.js + Express REST API	PostgreSQL Primary Database	Redis Cache + Sessions
Elasticsearch Full-Text Search	Bull + Redis Job Queues	Docker + K8s Containerization	AWS / GCP Cloud Hosting
OAuth 2.0 + JWT Authentication	SendGrid Transactional Email	Twilio SMS + Calls	Metabase Analytics Layer

16 Core Database Schema

Key tables and relationships

Table	Key Fields	Relationships
users	id, name, email, role_id, team_id	Belongs to role, team
leads	id, name, email, source, score, status, assigned_to	Belongs to user
contacts	id, first_name, last_name, email, account_id	Belongs to account, has many activities
accounts	id, name, website, industry, arr, health_score, owner_id	Has many contacts, opportunities, tickets
opportunities	id, name, amount, stage, close_date, account_id	Belongs to account; has many activities
activities	id, type, subject, due_date, related_type, related_id	Polymorphic: Lead / Contact / Deal

Table	Key Fields	Relationships
campaigns	id, name, type, status, open_rate, click_rate	Has many campaign_contacts
tickets	id, subject, priority, status, contact_id, account_id	Belongs to contact, account
workflow_rules	id, trigger_object, conditions (JSONB), actions (JSONB)	Evaluated on record change

17 Implementation Phases & Roadmap

Build in stages — ship value early

Phase 1 · Weeks 1–6

Foundation — Core CRM Objects

User auth, Leads module (CRUD, status, assign), Contacts & Accounts basic profiles, manual activity logging, basic list views with search and filter.

Phase 2 · Weeks 7–12

Sales Pipeline & Opportunities

Opportunities module with configurable stages, Kanban pipeline board, Lead → Opportunity conversion, basic dashboards, role-based access control.

Phase 3 · Weeks 13–18

Communications & Integrations

Gmail/Outlook two-way email sync, email open & click tracking, calendar sync, Twilio SMS + call logging, Zapier webhook endpoint.

Phase 4 · Weeks 19–24

Automation, Sequences & Marketing

Workflow builder (trigger → condition → action), email sequences, lead scoring engine, campaign module, custom report builder.

Phase 5 · Weeks 25–32

Support, Analytics & Scale

Support ticketing with SLA timers, NPS/CSAT survey automation, advanced forecasting, mobile app (React Native), public API + developer docs.

18 Best Practices

What separates good CRMs from great ones

Data Quality

- Enforce required fields at every stage gate
- Run deduplication jobs weekly
- Validate email and phone on entry
- Use picklists, not free text, for categorical fields
- Archive, don't delete — keep historical data
- Regular data enrichment via Clearbit / ZoomInfo

User Adoption

- Mobile-first design for field sales reps
- Minimize required fields — ask only what's needed
- Gamification: leaderboards and activity streaks
- Manager dashboards = accountability for data entry
- Onboarding checklist for new users
- In-app tooltips and contextual help

Common Failures to Avoid: Over-customizing before adoption, importing unclean data, too many required fields, giving everyone admin access, not training managers to inspect pipeline in the CRM, and letting data go stale by not integrating the tools reps use daily.

19 KPIs & Metrics to Track

How you know the CRM is delivering value

Category	Metric	Why It Matters
Revenue	MRR/ARR growth rate	Core business health
Revenue	Revenue by lead source	Allocate marketing budget
Revenue	Customer Lifetime Value (CLTV)	Pricing and retention investment
Revenue	Average Contract Value (ACV)	Segment targeting
Pipeline	Pipeline coverage (3–4x quota)	Forecast confidence
Pipeline	Stage-to-stage conversion rates	Identify bottlenecks
Pipeline	Average sales cycle length	Resource planning
Pipeline	Deal velocity	Rep effectiveness
Activity	Calls and emails per rep/day	Effort accountability
Activity	Time to first contact	Speed to lead performance
Activity	Task completion rate	Follow-through discipline
Customer Success	Churn rate	Retention health
Customer Success	Net Promoter Score (NPS)	Customer sentiment
Customer Success	Average ticket resolution time	Support quality
Customer Success	Renewal rate by segment	Expansion revenue risk